

Unlucky Sevens (Catan IMessage Game)

1. Vision & Objective

Create a full-length, async board-strategy game called Unlucky Sevens for iMessage that captures the core experience of Catan. The game should be playable end-to-end inside Messages bubbles (similar to Gamepigeon), support 3-4 players, and ultimately be modifiable to fit the groups' needs (time limits, house rules, etc.).

This document describes the user experience and specific design choices to make this digestible in an iMessage format. It is an evolving document, with the full Catan rules contained in the [appendix](#). It is not a technical specification, which is present in a different document.

2. Ideal-ish User Experience Flow

1. **Match Invitation** ⇒ iMessage bubble with “Join Unlucky Sevens” link.
2. **Generate Board** ⇒ this needs to be a fair board with a semi-even distribution across all resources. With that being said, this should be an RNG board with seeds, so there should be potential for there to be non-even boards as in the spirit of the game. Harbors, numbers, and resources distributions should all be placed at this stage.
 - *Robber should be on the desert to begin with
 - There should be a way to set fairness of the board, i.e. what sort of distribution does the group want for the game.
 - There should be a way for players to toggle certain settings:
 - i. Do not let 6s and 8s touch.
 - ii. Desert is only on coast or off-center
 - iii. Same resource as port
 - iv. Friendly Robber: a player must have greater than 2 VPs before a tile that they are on is targeted by the robber

3. **Initial Placement** (2 settlements + roads) via interactive board. Get resources from the second settlement. Settlements are done in a snake draft with a random order set at game start. The distance rule applies here, wherein any settlement can only be placed if there is no adjacent settlement or city (must be at least two edges away).
4. **Turn Screen** (single GameKit hand-off)
 - **Phase 0 - Play development card:** At any point during a player's turn, they can play exactly one development card from their hand that they acquired in a previous turn. If the player built a development card this turn, then they may not play that specific card.
 - i. **Knight:** move the robber to any tile, and if there are any players on that tile (excluding oneself), the steal pop-up allows the active player to select a target. The active player gets a card at random from the target.
 - ii. **Monopoly:** the active player can select a resource out of the 5 potential resources to acquire. All of the resources of that type that are in any of the other players' hands are given to the original player.
 - iii. **Year of Plenty:** the active player can choose any two resources and add them to their hand. These can be the same or different resources.
 - iv. **Victory Point:** this is a special development card. The player cannot play this card, but at any point if a victory point helps the player reach the desired amount of VPs to win the game (usually 10 in a standard game), this card is auto revealed and the player wins the game.
 - v. **Road builder:** the active player enters build mode (as described during the build phase) and may place up to two roads.
 - **Phase 1 – Roll:** dice $\Rightarrow$ resources auto-dealt based on either city (2 per touching tile) or settlement (1 per touching tile) existing.
 - i. **Robber Event:**
 1. When a 7 is rolled, players with >7 resource cards receive an auto-generated discard mini-turn.
 2. Active roller drags robber to any tile
 3. If there are any players on that tile (excluding oneself), the steal pop-up allows the active player to select a target.

- 4. The active player gets a card at random from the target.
 - ii. If a robber is on any tile, that tile does not generate resources even if the number for that resource is rolled
 - iii. If any resource runs out due to player payouts, no one gets that resource and it remains at the original count in the bank
- **Phase 2a – Trade:** optional Trade Drawer opens with the ability to compose Bank, Port, or player to player offers.
 - i. If P2P option, compose a message in chat for this. Another player can accept, but table talk before the turn is encouraged.
 - ii. In more competitive settings, there could be a timer before anyone can accept the trade to encourage table talk.

iii. **Peer-to-Peer/Peer-to-Port/Peer-to-Bank Trade Experience**

1. **Goals**

- a. Complete common trades in ≤ 3 taps.
- b. Keep Bank/Port trades distinct from P2P offers.
- c. Maintain hand secrecy so that no UI element reveals another player's resource counts or types. However, you should be able to see the total number of cards that a player has.
 - i. Clarification note: although the authoritative state contains the full hands, the UI must only reveal the opponent's hand count/size.

2. **Interaction Flow**

- a. The player taps the Trade drawer.
- b. Modal presents two tabs: Bank / Port and Players.
- c. Players tab lists opponents by avatar + name only (no resource icons or quantities). Each player should have the number of cards they have under their avatar + name.
- d. Offer Composer:
 - i. Two vertical stacks labelled Give and Get.

- ii. Each stack shows five resource chips with +/- steppers (or tap-to-cycle $0 \Rightarrow 1 \Rightarrow 2 \Rightarrow 3$).
 - iii. Players adjust counts to craft an offer, i.e. Give 2 wood $\Rightarrow$ 1 Brick
- e. Review Offer screen summarizes deal; player hits Send. A pop up saying that table talk in the group chat is encouraged before the trade offer should be visible here.
- f. Optionally, you can specify a recipient for the trade, but the trade would still be within the group chat.
- g. Recipient or table receives inline bubble:
 “Player” offers 2 wood for 1 brick — [Accept] [Decline].
 This would be through a message in the group chat.
 - i. Accept completes the trade immediately: both players’ hands update at once, and the offer is closed in the chat thread.
- h. The main player is the only one that can propose trades, so they can edit this based on table talk, before the trade is accepted.
- i. Trades must be atomic and completed before the End Turn phase, wherein they will go stale. They may also go stale after 12 hours of non-interaction, which can be configurable in the settings.

3. Port Quick Trades

- a. Ports are enabled by either a city or a settlement on the node that shares the port.
- b. On the Bank / Port tab, the composer auto-applies the best ratio (4:1 base, 3:1 generic, 2:1 specific) based on ports the active player owns.
- c. One-tap Confirm executes the trade.

- **Phase 2b – Build:** build Mode highlights legal nodes/edges, the player then places roads/settlements/cities. They can also optionally build a development card.
 - i. **Legality:**
 1. For settlements: there must be a settlement in hand, and there must be at least one road leading up to it with the distance rule applying here as well: there can be no settlement or city on an adjacent node (>1 edge away at least).
 2. For cities: there must be a city left in hand, and it must replace a settlement.
 3. For roads: there must be a road left in hand, and it must connect from a previous road or settlement/city.
 - ii. There should only be options to build items that are possible (i.e. a road can only be built if the player has a wood and a brick, etc.)
 - **Phase 2c – Win Condition Check:** If at any point during the build phase, the player has 10 VPs (or greater), they auto win, and the game goes to the Win Screen.
 - i. This phase includes checking for the largest army or longest road.
 - **Phase 3 – End Turn:** commits state & renders snapshot bubble to the chat.
5. **Win Screen** with confetti + shareable stats.
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2. Core Gameplay Components

Category	MVP	Phase 2+
Board	19-tile classic hex layout (randomized setup). Ports: ● 4 generic 3:1	Scenario maps, mini-boards, and map rerolls.

Category	MVP	Phase 2+
	5 2:1 corresponding to each resource	
Players	3–4 players with async turns.	Optional 2-player duels + AI fill-ins.
Resources	Wood, Brick, Wheat, Sheep, Ore,. There are 19 of each resource. Resources can be depleted.	TBA
Dice Roll	2× D6 roll; 7 triggers Robber.	Animated roll + shake-to-roll gesture.
Build Options	Road: 1 wood and 1 brick. Settlement: 1 wood, 1 brick, 1 sheep, 1 wheat. City: 3 ore, 2 wheat. Development cards: 1 sheep, 1 ore, 1 wheat.	TBA
Trading	Player ↔ Player & Bank (4:1, 3:1, 2:1).	Smart suggestions, trade history.
Robber	Drag robber to new tile, steal 1 card.	Optional seven = desert event trigger, Event animations..
Cities, Settlements, and Roads	Players have a piece supply of 5 settlements, 4 cities, and 15 roads.	
Largest Army and Longest Road	Largest Army: player has >2 knights played and the largest in the game. Gains 2 VPs.	

Category	MVP	Phase 2+
	Longest Road: player has >4 roads connected with no breakages (i.e. no other roads in the middle any other players). Gains 2 VPs.	
Development Cards	14 knight cards: move the robber. 5 victory point cards: automatically turn over when the user could make it to 10 Victory Points. 2 Road Builder cards: add two roads when played with road limitations. 2 Year of Plenty cards: add 2 resources to hand. 2 Monopoly Cards: retrieve all instances of resource from every player's hand Once depleted, there are no more development cards.	TBA
Victory	First to 10 VP.	TBA

3. Platform & Technology

- **Client:** iMessage App Extension written in SwiftUI.
 - **Multiplayer:** Fully in iMessage Turn-Based Matches.
 - **State Payload:** GameState $\Rightarrow$ JSON Data $\leq$ 64 KB per turn.
 - Authoritative GameState is stored in matchData and includes board, VP, and full hands (resource + dev) for each player. The UI enforces secrecy by showing opponents only handSize (total card count), not types or per-resource counts. Devices may cache decoded hands locally for performance/UI, but matchData is the source of truth.
 - **Graphics:** SpriteKit for hex board rendering $\Rightarrow$ snapshot PNG for bubble.
 - **Group Chat:** Runs inside an iMessage group thread. This is presented as an invite bubble that the user sends into a group thread (existing or newly created by the user).
 - **RNG Setup:** Dice and board setup is done using deterministic RNG. Record the seed for board and dice if needed.
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5. Milestones & Timeline

Phase	Deliverables
0 – Pre-Prod	Art style guide Updated PRD
1 – Engine	GameState model Board gen Turn logic
2 – Core UI	Hex renderer Tap-to-build Dice animation
3 – Trades, Robbers & Dev Cards	P2P trading UX

Phase	Deliverables
	Robber workflow Development Cards
4 – iMessage Glue	Bubble snapshot
5 – Beta & Polish	Crash handling Performance

6. Success Metrics

- $\geq 90\%$ games reach completion without state desync: ensuring that there are no issues with the game integrity as an iMessage Game
- Avg. active-player turn time ≤ 2 min: measure of an easy to understand UX flow
- Session resume lag ≤ 3 s
- Game satisfaction $\geq 4/5$: overall ratings on the game are high

7. Risks & Mitigations

Risk	Mitigation
Payload size limits	Delta encode board; compress snapshots.
Long async gaps	Push-notification nudges; optional 24 h turn timer.

8. Additional Requirements

8.1 Accessibility & Localization

- Dynamic Type & VoiceOver labels for all buttons and resource icons.
- Support for at least English in the first phase only (localized Strings file).

8.2 Telemetry & Analytics

- Anonymous event logging: roll distribution, average turn duration, trade frequencies.
- Opt-in crash reporting via Firebase Crashlytics (or Sentry).

8.3 Launch

- Cosmetics store (player-color skins) – non-gameplay.
- One-time unlock to host unlimited concurrent matches.

8.4 Privacy & Compliance

- No collection of personal data beyond Game Center ID.
 - GDPR/CPRA opt-out toggle.
 - Age 13+ rating (no in-app chat beyond standard iMessage).
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9. Open Questions (as they come up)

1. Are we handling authorization in this initial design?
 - a. No, as authorization complexities come up, they may be addressed, but assume that they are built in line with Apple standards
2. Are we handling hand secrecy + matchData?
 - a. No, assume that all people playing are friends/do not want to cheat. This may be addressed in the future, but it is more important to have a launch first. The game is social/group chat game, so if a player is consistently cheating, they can be caught and shamed socially.

Appendix

Catan Rules

- Objective
 - Be the first player to reach 10 Victory Points (VP) on your turn (immediately wins).
- Setup
 - Build the island with 19 hex tiles (resources: wood, brick, wheat, sheep, ore; 1 desert).
 - Place number tokens (2–12) on non-desert tiles; robber starts on the desert.
 - Each player, in snake order ($1 \rightarrow N$, $N \rightarrow 1$), places:
 - 1 settlement on any vertex + 1 connected road.
 - Then a second settlement + road (reverse order).
 - Starting resources: collect 1 of each resource from the second settlement's adjacent tiles.
- Turn Structure (one handoff)
 - Play 1 development card (optional; restrictions below) at any time.
 - Roll 2d6:
 - If 7: see “Robber”.
 - Else: all tiles with that number produce (see Production).
 - Trade (optional): with players and/or Bank/Ports.
 - Build/Buy (optional): roads, settlements, cities, development card.
 - End turn.
- Production
 - For each producing tile:
 - Adjacent settlement owners gain 1 of that resource.
 - Adjacent city owners gain 2 of that resource.
 - Robber-blocked tiles produce nothing.
- Trading

- Domestic trade: propose any deal with any players; all terms allowed.
- Bank trade: 4:1 of one resource for any 1.
- Ports (if you own a settlement/city at that port):
 - 3:1 (generic) or 2:1 (specific resource) with the bank.
- Building (costs & rules)
 - Road = 1 wood + 1 brick
 - Place on an edge connected to your network (road/settlement/city).
 - Settlement = 1 wood + 1 brick + 1 wheat + 1 sheep
 - Must be at a vertex connected to your road and obey the distance rule: no other settlement/city on adjacent vertices (1 edge away).
 - City (upgrade) = 2 wheat + 3 ore
 - Replace your settlement at that vertex (remove the settlement piece).
 - Piece limits: per player 15 roads, 5 settlements, 4 cities; you cannot build beyond your supply.
- Robber (rolling a 7 or playing Knight)
 - Discard: any player with >7 resource cards discards half (rounded down).
 - Move robber to any tile; that tile is blocked until robber moves again.
 - Steal 1 random resource from one opponent with a settlement/city on that tile.
- Development Cards (buy: 1 wheat + 1 sheep + 1 ore)
 - Types:
 - Knight: move robber + steal 1 (counts toward Largest Army).
 - Road Building: place 2 roads for free (obey placement rules).
 - Year of Plenty: take any 2 resources from the bank.
 - Monopoly: choose a resource; all other players give you all of that resource.
 - Victory Point: +1 VP, kept hidden until it is possible to reach 10 points with the VPs to win. If so, those are automatically turned over.
 - Play limits:
 - 1 dev card per turn.
 - You cannot play a dev card on the same turn you bought it (except you may reveal VP to win).

- Dev cards are not tradable.
- Special Awards
 - Longest Road (2 VP): longest continuous road of ≥ 5 edges; can change hands.
 - An opponent's settlement breaks your continuity; your own does not.
 - Largest Army (2 VP): first to have played ≥ 3 Knights; can change hands.
- Other Rules & Clarifications
 - Bank is finite: if the bank runs out of a resource, no one can take more of it until returned. This means that if a bank resource runs out as players are trying to collect it (i.e. 3 wheat left, but all 4 players need to pick up a wheat), then no one gets to pick up the resource and all players must return the wheat until it is possible to pick up the resource without it running out.
 - Hand size: no max hand size (only the robber's discard on 7).
 - Build timing: you may build/trade in any order after rolling; place pieces immediately upon paying.
- Winning: the moment you reach 10+ VP on your turn, you win immediately.